

UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE AND ENGINEERING

TEST 2, AUGUST 2026

DURATION: 3 hours

MIE 1626 H — Data Science Methods and Statistical Learning

Calculator Type: None

Exam Type: Open: reference sheet

Aids allowed: the reference sheet distributed with the test

Examiner(s): M. Guerzhoy

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Family Name(s):

Given Name(s):

*Do **not** turn this page until you have received the signal to start.*
In the meantime, please read the instructions below carefully.

This test paper consists of 6 questions on 15 pages (including this one), printed on both sides of the paper. *When you receive the signal to start, please make sure that your copy is complete, and fill in the identification section above.*

Answer each question directly on this paper, in the space provided. Use the pages at the end of the test for extra space. If you use extra pages, indicate that you have done so in the space under the question.

You can answer all 6 questions. Your exam score will be computed using the best 5 questions.

Write up your solutions carefully! Comments are *not* required to receive full marks. However, they may help us mark your answers, and part marks *might* be given for partial solutions with comments clearly indicating what the missing parts should accomplish.

Use the **R** programming language. You may assume the **tidyverse** (and, where relevant, **lme4**) is loaded and that the data frames named in each question are already available. Verb signatures (**group_by**, **summarize**, **filter**, **lm**, **glm**, **lmer**, **sample**, **replicate**, **plogis**, **fisher.test**, ...) are on the reference sheet, so exact syntax need not be memorized and minor syntax slips are not penalized. You are *not* required to write **ggplot** code; plotting questions ask you to *describe* or *read* a plot.

When you are asked to write code, *no* error checking is required: you may assume that all input and argument values are valid, except where explicitly indicated otherwise.

MARKING GUIDE

Q1: _____/ 20

Q2: _____/ 20

Q3: _____/ 20

Q4: _____/ 20

Q5: _____/ 20

Q6: _____/ 20

TOTAL: _____/120

Question 1. [20 MARKS]

The `gapminder` data frame has one row per country per year. Its first rows for 2007 look like this:

	country	continent	year	lifeExp	pop	gdpPercap
1	Canada	Americas	2007	80.653	33390141	36319.24
2	Norway	Europe	2007	80.196	4627926	49357.19
3	Kuwait	Asia	2007	77.588	2505559	47306.99
4	Australia	Oceania	2007	81.235	20434176	34435.37
5	Nigeria	Africa	2007	46.859	135031164	2013.98

The `continent` column takes the values Africa, Americas, Asia, Europe, Oceania.

Write R code that returns, **for each continent, the country with the highest gdpPercap in 2007**. Your code should return one row per continent identifying the country.

Question 2. [20 MARKS]

You flip a bent coin $n = 10$ times, independently, and observe $k = 3$ heads. Let p be the (unknown) probability of heads.

Part (a) [9 MARKS]

Write R code that finds the maximum-likelihood estimate $\hat{p}$ of p by **grid search** — that is, by evaluating the likelihood (or log-likelihood) over a grid of candidate values of p and taking the best one. State the likelihood you are evaluating.

Part (b) [11 MARKS]

Now write code to estimate p **and its uncertainty** using a **Bayesian** approach, so that your answer corresponds to a region containing the true p with 95% probability. State your **prior**, say what the **posterior** is over, and give R code (a grid is fine) that returns a 95% credible interval for p .

Question 3. [20 MARKS]

A logistic-regression model outputs $\hat{p}$, the estimated probability that a credit-card transaction is **fraudulent**. For each transaction you must decide whether to **block** it. Blocking a *legitimate* transaction costs you \$5 (an annoyed customer); letting a *fraudulent* transaction through costs you \$200. You block whenever $\hat{p} > t$.

Part (a) [12 MARKS]

Write the **expected cost** of a single transaction under each action (block / don't block), as a function of $\hat{p}$, and derive the threshold t^* at which the two actions break even.

Part (b) [8 MARKS]

The model is well-calibrated, but the \$200 figure is only a rough guess. Explain, in one or two sentences, which way t^* moves if the true cost of missed fraud is actually *higher* than \$200, and why.

Question 4. [20 MARKS]

For the Minnesota radon data, one fits `lmer(log_radon ~ floor + (1 | county))`, where `floor` is 1 for a first-floor measurement and 0 for a basement measurement. The fit gives a `floor` coefficient of about -0.6 , and the estimated standard deviation of the county intercepts is about 0.3.

Part (a) [10 MARKS]

In English, explain what the `floor` coefficient and the county-intercept standard deviation mean. That is, is a manager asks you what the numbers

Part (b) [10 MARKS]

You must predict radon for a house in a county that was **not** in the data. What is your best estimate of that county's intercept, and why?

Question 5. [20 MARKS]

Graduate admissions decisions are made roughly on the sum **GPA** + **GRE**: an applicant is admitted if that sum is high enough. In the **general applicant pool**, **GPA** and **GRE** are *positively* correlated.

Part (a) [9 MARKS]

Among **admitted** students only, what do you expect the sign of the correlation between **GPA** and **GRE** to be, and why? (You do not need to give a formula, just explain in words/pictures)

Part (b) [11 MARKS]

Draw the three-node DAG for `GPA`, `GRE`, `LETTERQUALITY` and `Admit`, and propose a formula for $P(\textit{Admit} = 1 \mid \textit{GPA}, \textit{GRE}, \textit{LETTERQUALITY})$ that is consistent with the DAG. Justify one of the arrows in your DAG and the lack of one of the arrows in your DAG that could be there but you chose not to include.

Question 6. [20 MARKS]

You have scraped about 3000 posts from the r/UofT subreddit and want to answer the question “**why** are students (un)happy?”

Part (a) [5 MARKS]

Your first instinct is to report “62% of posts are negative.” Explain why that single number is an unsatisfying answer, and say what would be more useful.

Part (b) [5 MARKS]

Contrast **zero-shot** and **few-shot** prompting for labelling a post’s theme, and say when few-shot helps.

Part (c) [5 MARKS]

You then label themes by clustering **embeddings** of the posts. What is an embedding here, and why cluster embeddings rather than cluster on shared keywords?

Part (d) [5 MARKS]

You have the model label all 3000 posts as “complaint about workload” or not. How would you decide whether to **trust** those labels, and what single number would you report?

*Use the space on this “blank” page for scratch work, or for any solution that did not fit elsewhere.
Clearly label each such solution with the appropriate question and part number.*

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